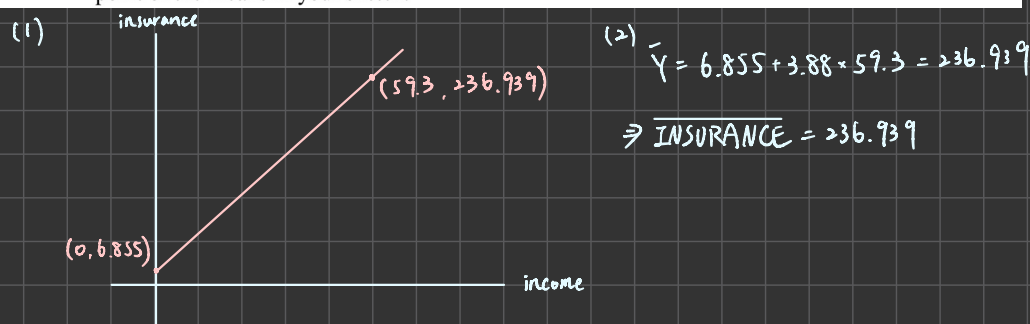


3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880 INCOME$$

(se) (7.383) (0.112)

- a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of *INCOME* = 59.3. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.



- b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

$$se(\hat{\beta}_1) = 0.112 \quad df = 18 \quad t_{0.025, 18} \approx 2.101$$

$$\text{point estimate} : 3.88 \times 1000 = 3880$$

$$95\% \text{ CI} : \hat{\beta}_1 \pm t_{0.025, 18} \cdot se(\hat{\beta}_1)$$

$$\Rightarrow 3.88 \pm 2.101(0.112)$$

$$\Rightarrow 3.88 \pm 0.2353$$

$$\Rightarrow (3.6447, 4.1153)$$

$$\times 1000 \Rightarrow [3644.7, 4115.3]$$

We estimate that for each additional \$1000 of household income, the average amount of life insurance held increases by about \$3880, with a 95% confidence interval from \$3644.7 to \$4115.3.

- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746.

$$income = 100$$

$$\hat{y}_0 = 6.855 + 3.88(100) = 394.855$$

$$Var(\hat{y}_0) = Var(\hat{\beta}_0) + X_0^2 Var(\hat{\beta}_1) + 2X_0 Cov(\hat{\beta}_0, \hat{\beta}_1)$$

$$= 9.383 + 100^2(0.112^2) + 2(100)(-0.746) = 30.7487$$

$$se(\hat{y}_0) = \sqrt{30.7487} = 5.545$$

$$t_{0.005, 18} = 2.878$$

$$99\% \text{ CI} = 394.855 \pm 2.878(5.545)$$

$$= 394.855 \pm 15.76$$

$$= [378.9, 410.82]$$

- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

$$\begin{cases} H_0: \beta_2 = 5 \\ H_a: \beta_2 \neq 5 \end{cases} \quad \hat{\beta}_2 = 3.88 \quad se(\hat{\beta}_2) = 0.112 \quad t = \frac{\hat{\beta}_2 - 5}{0.112} = \frac{3.88 - 5}{0.112} = -10$$

$$t_{\alpha/2, 18} = 2.101$$

$$\Rightarrow -10 < -2.101 \Rightarrow \text{Reject } H_0$$

The data do not support the claim that each additional \$1000 of income increases the average amount of life insurance held by \$5000.

- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

$$\begin{cases} H_0: \beta_2 = 1 \\ H_a: \beta_2 > 1 \end{cases} \quad t = \frac{3.88 - 1}{0.112} = 25.714 \quad t_{\alpha, 18} = 2.552$$

$25.714 > 2.552 \Rightarrow \text{Reject } H_0$. There is strong evidence that the slope greater than one.